

Imperial College London

BSc/MSci EXAMINATION May 2022

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Monday, 30th May 2022: 14:00 to 16:00

*The paper consists of **three** questions – please answer **all three**.
A table of Laplace transforms is included at the end of the paper.*

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

At the top of each page of your answers, write your CID number, module code, question number and page number. Scan and upload your answers to the Turnitin dropboxes as described in the guidance documents in the Blackboard module for this exam. Upload each answer to the dropbox provided for that specific question.

For each answer you should prepare a coversheet which should be the first page of your scanned answer. The coversheet should contain the following:

- your CID
- module name and code
- the question number
- the number of pages in your answer

You should not write your name anywhere on your answers.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. A photomultiplier tube (PMT) set up for photon counting generates a voltage $v(t)$ across a load resistor $R_L = 50 \, \Omega$ connected between its anode and ground.

(i) The PMT response to a single photoelectron is approximately Gaussian with FWHM = 10 ns and peak amplitude of 5 mV. Calculate the PMT gain. [8 marks]

(ii) The PMT is now connected to a fast amplifier by a 20 m long coaxial cable with characteristic impedance $Z_0 = 50 \, \Omega$ and capacitance per unit length $C' = 100 \, \text{pF/m}$.

(a) Define 'characteristic impedance' for a generic transmission line and indicate how this is calculated for a lossless coaxial cable. [5 marks]

(b) Calculate the signal delay in this 20 m long cable. [5 marks]

(c) The amplifier has input impedance $Z_i = 75 \, \Omega$; characterise the reflected signal (amplitude and polarity) arising from this impedance mismatch for single photoelectron signals. How does this change the single photoelectron response? [5 marks]

(iii) The amplifier has a voltage gain of 20 dB and a constant noise spectral density $v_n = 1.5 \, \text{nV}/\sqrt{\text{Hz}}$ (referred to the input) over its 300 MHz bandwidth (−3 dB).

(a) Calculate the amplitude of the single photoelectron signal at the output. [5 marks]

(b) What is the signal-to-noise ratio (in dB) for the single photoelectron peak amplitude? [5 marks]

(c) Choose the sampling speed and resolution of a suitable digitiser for this system, considering a full scale of 0.1 V. Briefly justify your answer. [7 marks]

[Total 40 marks]

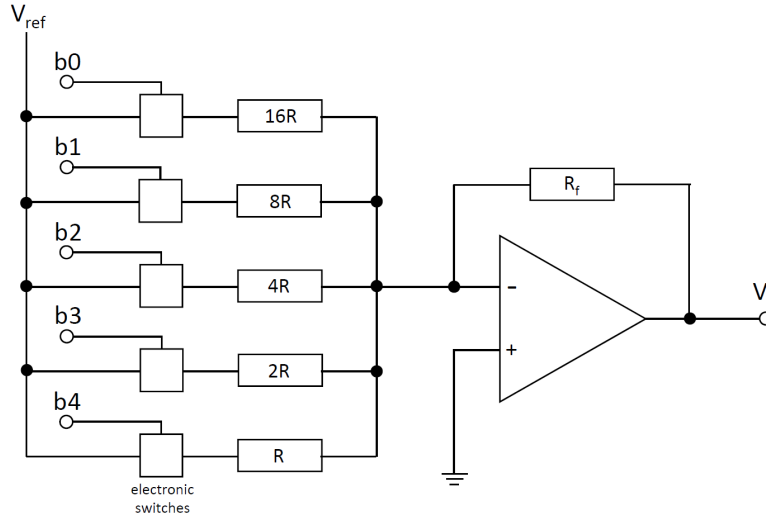
Useful information.

Electron charge $e = 1.602 \times 10^{-19} \, \text{C}$.

Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{t^2}{2\sigma^2}} dt = \sqrt{2\pi}\sigma$$

2. Consider the 5-bit Digital-to-Analogue Converter (DAC) shown in the figure, which employs an operational amplifier, initially considered as an ideal device. The DAC converts a binary number $b_4 b_3 b_2 b_1 b_0$, where each bit is a binary value (0 or 1), into an analogue voltage V_o . The bits correspond to voltages of 0 V for a logical '0' and $V_{ref} = -1$ V for a logical '1'.



- (i) Briefly state the properties of the ideal operational amplifier *and* give the two 'golden-rules' used to simplify the analysis of circuits containing them. [5 marks]
- (ii) Show that the DAC output can be written as a function of the binary input as:

$$V_o = -V_{ref} \frac{R_f}{R} \left(b_4 + \frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right).$$
 [6 marks]
- (iii) Derive the value of the feedback resistor R_f for the DAC to cover a maximum output range of 5 V. Then indicate the voltage corresponding to the least significant bit and the dynamic range of the DAC in dB. [5 marks]
- (iv) In some op-amp circuits a resistor (R_1) is added between the non-inverting input and ground in order to compensate for the finite impedance of the inputs. Explain how this would work and calculate the resistor value for this design. [6 marks]
- (v) Consider that a sine wave with period $200 \mu s$ is digitised into a 5-bit binary number and fed into the DAC input; the clock time per sample is $10 \mu s$, during which time the input does not change. The amplitude is such that the wave uses most of the full scale available to the DAC. Sketch one period of the DAC output in one plot and the frequency spectrum of the DAC output in another. Explain the most relevant features in each plot, using numerical values where possible. [8 marks]

[Total 30 marks]

3. (i) Give the integral definition of Laplace transform; then derive its time delay property: if a signal $g(t)$ is equal to another signal $f(t)$ delayed by time T , i.e.

$$g(t) = \begin{cases} 0 & 0 \leq t < T \\ f(t - T) & t \geq T \end{cases}$$

show that its Laplace transform is

$$G(s) = e^{-sT} F(s).$$

[5 marks]

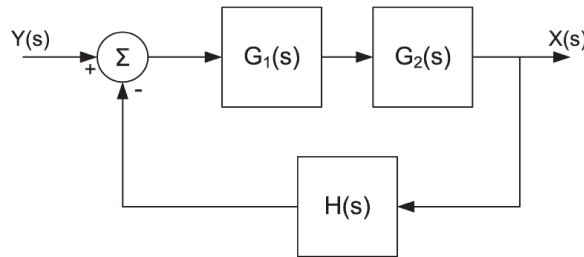
- (ii) Sketch the Bode plot for a lossless transmission line with time delay T .

[5 marks]

- (iii) Consider the negative feedback system represented in the figure below. Show that the system has the following transfer function:

$$T(s) = \frac{G_1(s)G_2(s)}{1 + H(s)G_1(s)G_2(s)}$$

[5 marks]



- (iv) Evaluate the transfer function for the following case:

$$\begin{aligned} h(t) &= \delta(t) \\ g_1(t) &= e^{-2t} \\ g_2(t) &= u(t) \end{aligned}$$

[5 marks]

- (v) For a unit step $u(t)$ applied to the input at $t = 0$, find an expression for the output $x(t)$ as a function of time for $t > 0$.

[5 marks]

- (vi) Considering the similarity between the transfer function and the frequency response of a system, calculate how this system would respond to a sinusoidal signal with angular frequency ω .

[5 marks]

[Total 30 marks]

Useful information:

In addition to the Laplace transforms at the end of this paper you may need the following inverse transform:

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s + a)^n} \right\} = \frac{t^{n-1} e^{-at}}{(n-1)!}$$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Figure 1: Laplace transforms.